

# **Engineering Reboot 2026**

## **Week-02 Engineering Lab Workbook v3.5**

### **MASTER WORKBOOK - LOCK-IN EDITION**

This workbook is intended to be a self-contained execution package. All exercises include file locations, prompts, expected output, test cases, screenshots, notes, journals, and Git workflow.

# Day 06 - Conditional Logic

## Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible

## Exercise / Deliverable: age\_classifier.py

Location: Week-02\Day-06\Exercises\age\_classifier.py

Required Prompt(s): Input age:

Expected Result(s): 5 -> Classification: Child; 13 -> Teen; 35 -> Adult; 70 -> Senior

## Exercise / Deliverable: temperature\_advisor.py

Location: Week-02\Day-06\Exercises\temperature\_advisor.py

Required Prompt(s): Input temperature:

Expected Result(s): 40 -> Wear Jacket; 65 -> Comfortable; 90 -> Warm Weather

## Notes File

Week-02\Day-06\Notes\Day-06-Notes.md

## Journal File

Week-02\Day-06\Journal\Day-06-Journal.md

## Git Commit Message

Day 06 - Completed exercises and documentation

## Completion Checklist

- ☐ All exercises complete
- ☐ All test cases executed
- ☐ Screenshots captured
- ☐ Notes completed
- ☐ Journal completed
- ☐ Git commit completed
- ☐ Git push completed

# Day 07 - Boolean Logic

## Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible

## Exercise / Deliverable: eligibility\_checker.py

Location: Week-02\Day-07\Exercises\eligibility\_checker.py

Required Prompt(s): Input age:, Input citizenship status (Y/N):

Expected Result(s): 25,Y -> Eligible YES; 16,Y -> NO; 30,N -> NO

## Exercise / Deliverable: building\_access\_control.py

Location: Week-02\Day-07\Exercises\building\_access\_control.py

Required Prompt(s): Badge Present (Y/N), Manager Override (Y/N)

Expected Result(s): Y,N -> Granted; N,Y -> Granted; N,N -> Denied

## Notes File

Week-02\Day-07\Notes\Day-07-Notes.md

## Journal File

Week-02\Day-07\Journal\Day-07-Journal.md

## Git Commit Message

Day 07 - Completed exercises and documentation

## Completion Checklist

- ☐ All exercises complete
- ☐ All test cases executed
- ☐ Screenshots captured
- ☐ Notes completed
- ☐ Journal completed
- ☐ Git commit completed
- ☐ Git push completed

# Day 08 - Loops

## Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible

## Exercise / Deliverable: counting\_program.py

Location: Week-02\Day-08\Exercises\counting\_program.py

Required Prompt(s): No input

Expected Result(s): Display 1 through 20

## Exercise / Deliverable: even\_number\_generator.py

Location: Week-02\Day-08\Exercises\even\_number\_generator.py

Required Prompt(s): No input

Expected Result(s): Display 2 through 20 even numbers

## Exercise / Deliverable: sum\_calculator.py

Location: Week-02\Day-08\Exercises\sum\_calculator.py

Required Prompt(s): No input

Expected Result(s): Sum = 5050

## Notes File

Week-02\Day-08\Notes\Day-08-Notes.md

## Journal File

Week-02\Day-08\Journal\Day-08-Journal.md

## Git Commit Message

Day 08 - Completed exercises and documentation

## Completion Checklist

- ☐ All exercises complete
- ☐ All test cases executed
- ☐ Screenshots captured
- ☐ Notes completed

- ☐ Journal completed
- ☐ Git commit completed
- ☐ Git push completed

# Day 09 - Loop Practice

## Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible

## Exercise / Deliverable: square\_pattern.py

Location: Week-02\Day-09\Exercises\square\_pattern.py

Required Prompt(s): Input square size: 5

Expected Result(s): 5x5 star square

## Exercise / Deliverable: right\_triangle\_pattern.py

Location: Week-02\Day-09\Exercises\right\_triangle\_pattern.py

Required Prompt(s): Input rows: 5

Expected Result(s): Right triangle pattern

## Exercise / Deliverable: inverted\_triangle\_pattern.py

Location: Week-02\Day-09\Exercises\inverted\_triangle\_pattern.py

Required Prompt(s): Input rows: 5

Expected Result(s): Inverted triangle pattern

## Exercise / Deliverable: multiplication\_table.py

Location: Week-02\Day-09\Exercises\multiplication\_table.py

Required Prompt(s): Input multiplier: 7

Expected Result(s): 7 x 1 through 7 x 10

## Exact Pattern Outputs

Square Pattern Input 5

```
*****
*****
*****
*****
*****
```

Right Triangle Input 5

```
*  
**  
***  
****  
*****
```

Inverted Triangle Input 5

```
*****  
****  
***  
**  
*
```

## Notes File

Week-02\Day-09\Notes\Day-09-Notes.md

## Journal File

Week-02\Day-09\Journal\Day-09-Journal.md

## Git Commit Message

Day 09 - Completed exercises and documentation

## Completion Checklist

- ☐ All exercises complete
- ☐ All test cases executed
- ☐ Screenshots captured
- ☐ Notes completed
- ☐ Journal completed
- ☐ Git commit completed
- ☐ Git push completed

# Day 10 - Menu Driven Calculator Project

## Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible

## Exercise / Deliverable: menu\_driven\_calculator.py

Location: Week-02\Day-10\Projects\menu\_driven\_calculator.py

Required Prompt(s): Menu choices 1-5

Expected Result(s): Add, Subtract, Multiply, Divide, Exit

## Functional Requirements

- FR-01 Display Menu
- FR-02 Addition
- FR-03 Subtraction
- FR-04 Multiplication
- FR-05 Division
- FR-06 Division By Zero Handling
- FR-07 Exit Program

Division By Zero Expected Output:

Error: Division by zero is not allowed.

## Notes File

Week-02\Day-10\Notes\Day-10-Notes.md

## Journal File

Week-02\Day-10\Journal\Day-10-Journal.md

## Git Commit Message

Day 10 - Completed exercises and documentation

## Completion Checklist

- ☐ All exercises complete
- ☐ All test cases executed
- ☐ Screenshots captured
- ☐ Notes completed



- ☐ Journal completed
- ☐ Git commit completed
- ☐ Git push completed

## **Week-02 Completion Checklist**

- ☐ Day-06 Complete
- ☐ Day-07 Complete
- ☐ Day-08 Complete
- ☐ Day-09 Complete
- ☐ Day-10 Complete
- ☐ Repository Up To Date
- ☐ Ready For Week-03